ANALYSTLAB AFRICA

Machine Learning Internship Programme

Business Understanding Report

Field	Detail
Project	Customer Churn Prediction Project
Client	ABC Communications Ltd
Week	Week 1 — Machine Learning Problem Framing & Data Understanding
Role	Junior Machine Learning Engineer, AnalystLab Africa Consulting
Dataset	Telco Customer Churn Dataset (Kaggle, blastchar/telco-customer-churn)

1. Business Scenario

ABC Communications Ltd is a telecommunications provider offering phone, internet, and value-added services (streaming, security, backup, tech support) on month-to-month, one-year, and two-year contracts. Like most subscription businesses, ABC Communications loses a portion of its customer base every billing cycle. Acquiring a new customer is significantly more expensive than retaining an existing one, so the client wants to move from a reactive retention posture (acting only after a customer cancels) to a proactive one (intervening before a customer decides to leave).

2. Business Problem

ABC Communications Ltd needs a way to identify, in advance, which currently active customers are at high risk of churning (cancelling their service), so that the retention and customer-success teams can target them with tailored interventions — discounts, service upgrades, proactive support, or contract incentives — before they leave.

Framed as a machine learning problem: given a customer's demographic profile, account details, and service subscriptions, predict whether that customer will churn.

3. Business Questions

Question	Answer
What business problem should the model solve?	Predict which active customers are likely to churn so retention teams can intervene proactively rather than reactively.
What is the target variable?	Churn — a binary label (Yes / No) indicating whether the customer left the company.
Which input features should be used?	All customer attributes except the identifier: demographics (gender, SeniorCitizen, Partner, Dependents), account information (tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges), and subscribed services (PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies).
Is this classification or regression?	Classification — specifically binary classification, since the target has two discrete classes (Yes/No).
Which evaluation metrics should be used?	Precision, Recall, F1-score, and ROC-AUC. Accuracy alone is misleading here because the classes are imbalanced (~73.5% No vs. ~26.5% Yes).
Which algorithms are suitable?	Logistic Regression as an interpretable baseline; Random Forest and Gradient Boosting (XGBoost/LightGBM) as stronger candidates for capturing non-linear relationships. Full rationale is in the Machine Learning Proposal.

4. Objectives

- Understand and document the business problem from a machine learning perspective.
- Identify the target variable and the input features available in the dataset.
- Confirm the problem is a classification task, not a regression task.
- Inspect the dataset for structure, data quality issues, and class balance.
- Recommend suitable algorithms, evaluation metrics, and a preprocessing strategy for the modelling phase in the coming weeks.

5. Success Criteria

This phase of the project (Week 1) is considered successful if the following are clearly documented and understood before any model is built:

- A precise, unambiguous statement of the target variable and prediction task.
- A complete inventory of usable features, with data types and known quality issues.
- An evaluation strategy appropriate to the class imbalance present in the data.
- A shortlist of algorithms appropriate to a tabular, mixed-type binary classification problem.

6. Stakeholders

Stakeholder	Interest
ABC Communications Ltd (Client)	Reduce customer attrition and protect recurring revenue.
Retention / Customer Success Team	Needs a ranked, actionable list of at-risk customers and the drivers behind each prediction.
AnalystLab Africa Consulting	Delivers an accurate, well-documented, and reproducible churn model.
Junior ML Engineer (this role)	Frames the problem correctly, inspects the data, and proposes a sound modelling approach.